Project 03

Fresh Loop

Service Design / Sustainability / UI Design /Social Innovation

BRIEF

I observed that a significant amount of the **produce is underutilized.** So I envisioned FreshLoop—a community-based and digitally enabled sustainable system that **circulates surplus produce** locally and turns food waste into **reusable resources**, creating a smart, green loop.

INDIVIDUAL PROJECT

DURATION

Sep. 2025 – Nov. 2025



INSPIRATION

Fresh Produce Waste

Global Produce waste reaches **1.6 billion tons** each year, generating about **3.3 billion tons of carbon emissions.**

The average loss rates in 2025

Vegetables	*19-29%*
Fruits	*14-22%*

Approximately 10% of produce is lost in the **supply chain**, followed by another 17% during **home storage**.

Wholesale-Style Food Purchases

Consumers tend to buy **larger quantities of produce** and **stock up on them for later use.**

38.78 BILLION *In 2024* → **42 BILLION** *In 2025*

Safety and Quality of Produce

Residents are increasingly concern **produce safety & quality.**



Expressed safety concerns about market



Have started growing their own or buying from local sources.

Urban Farm

Homegrown produce is rapidly gaining popularity, with household gardens typically set up in **courtyards, on open terraces, or rooftop patios.**



Environmental Issues from Surplus Produce

Surplus produce directly cause **local environmental problems,** such as rotting leaves and peels piling up on streets, attracting pests, and blocking drains etc.

Odor *Blockage* *Pollution* *Pests* *Bacteria*

0.14 KG *Per person per day*

Eco-Friendly Treatment Methods

To reduces waste and minimizes environmental impact, contributes to sustainable resource use, the new method of recycle should be concerned.

— Composting

*Small-scale composting often uses **microbial decomposition**, with simple procedures and minimal environmental impact.*



— Materials Recycling

***Surplus produce and kitchen waste** can be converted into useful **bio-based materials** through proper processing.*



The Community

People typically live in close-knit, community-based settings where most residents are acquainted and share a deep level of trust.

Case Study

Many groups now partner with stores to sell "food boxes" of near-expiration food，mostly ready-to-eat items—to help reduce waste.

Too Good To Go — Copenhagen



*TGTG, Founded in Denmark, sells food that are nearing expiration dates and **has saved over 350 million meals** by the end of 2025.*

"XiShi" Food Magic Bag — Beijing



*XiShi, Founded in China, mostly provided by bakeries. They can be **reserved in advance**, but cannot be chosen.*

Trust Challenges in Redistribution

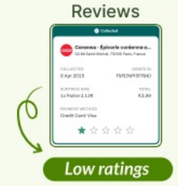
Consumers' concerns:

Food safety *Seller credibility* *Reviews*

*Many consumers fear—or have received **spoiled or poor-quality food:***





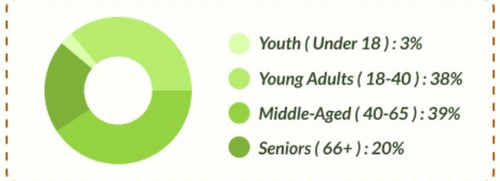


Low ratings

DESK RESEARCH

Demographics of Urban Farm Enthusiasts

Age distribution



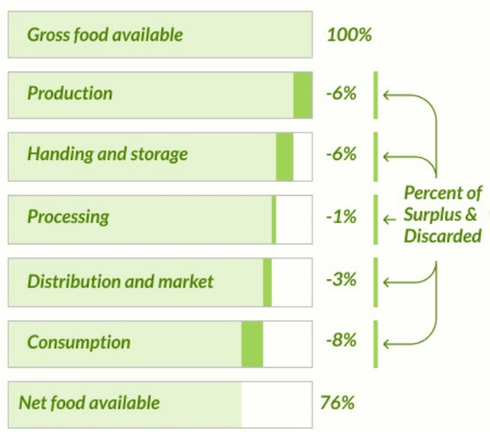
Occupational status



*The proportion of residents with urban farms is growing rapidly, expanding **from retired groups to younger urban populations.***

Loss or Waste at Each Stage

Approximately **24% of fresh produce is wasted** across the supply chain.



© From World Resources Institute

*The loss during production and processing is inevitable, so the primary focus should be on **storage, distribution and consumption process.***

The Redistribution and Recycling Windows for Surplus and Discarded Produce



2 New Bio-based Materials Recycling



Highly Perishable *Short-Term* *Medium-Term* *Long-Term*

1-3 days | 3-5 days | 1-2 weeks | Weeks

***Surplus food** can get redistribution first and then be converted into useful **bio-based materials** and **bio-fertilizer** through proper processing.*

1 Redistribution



*Excess food from bulk purchases or garden harvests can be **circulated within the community.***



Context Summary

Waste of fresh ingredients | *Pollution from Discarded Food* | *Technology of Recycling and Reuse* | *Household Garden*

*The **high loss rate of fresh ingredients** results in substantial waste, while **food discards** cause urban pollution and intensify carbon emissions and environmental strain. Establishing a more **efficient and higher-utilization cycle connecting food, humans, and the environment** is imperative.*

FIELD RESEARCH

Community Observation and Interviews



8 Inability to **accurately assess** food freshness.



9 Daily leftovers are small and diverse, complicating disposal.



10 Bulk purchases **do not fully match** actual needs.



1 ***Erratic yields:*** *" We are often caught in a cycle of no harvest or overwhelming surplus."*

Cause surplus

2 *Poorly planned crop varieties lead to alternating periods of **surplus and scarcity.***



Miss Lin
Studio apartment



Mrs.Zhou
House with a yard



3 *Lots of rotten or expired ingredients **get mixed up.***

4 *Food naturally decomposes will **emits methane**, intensifying the **greenhouse effect.***



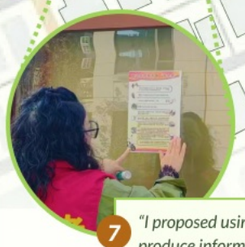
Mr.Wang
High-rise residence

*Leaving surplus fresh ingredients around leads to **pollution** in the neighborhood.* 5



*A **bountiful harvest** is often **wasted** or just shared among **family and friends.*** 6

Lack of effective communication



7 *"I proposed using the **bulletin board as a shared platform** for produce information to improve communication efficiency."*

Buildings
Community boundaries
Shared community space

Persona



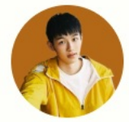
Miss Lin
28 years old
Liveing by herself as an apartment resident, a small-scale grower of vegetables.

- Sustainable Mindset
- Reusing handicraft
- Balcony Gardener
- Delivery App User



Mrs.Zhou
62 years old
Retired; owning a yard and growing large amounts of fresh produce.

- Significant free time
- High-Yield Grower
- Self-Sufficient Gardener
- WeChat Sharer



Mr.Wang
35 years old
A family of four living in a high-rise apartment with no space for gardening.

- Price-Sensitive
- Strongly Anti-Waste
- stockroom-style supermarket buyer

User Journey Map

Stages	Grow Fresh produce	Market and Storage	Consumption and Distribution	Waste disposal
Actions	Choose varieties; Balcony planting; Garden growing	Fertilizing; Harvesting; Online buying; supermarket stocking-up	Consume	Share online; Find peers; Surplus food; Inedible parts; Seek recycling options; Discard; Sell at a stall; As a gift; Reduce economic loss
Thinking	Costly & hassle; But fulfilling.	Satisfying; Inconveniently distant.	Enjoy fresh food; Pity: too much wasted	Can be reduced; How to do it ?
Feeling				
Painpoints	• ***The cost** of growing vegetables **exceeds** the cost of buying them.* • ***Simultaneous maturity** of food.*	• ***Be Skeptical** of store-bought Organic Labels.* • ***Too far** to deliver or buy.*	• *Neighbor network **cannot absorb surplus.*** • ***Overstocking and short shelf life** lead to predictable fresh food waste.*	• *Daily reusable materials are **too scattered to use.*** • ***Strong guilt** over food waste.*
Needs	• *Obtain **low-cost** planting materials & fertilizers* • *Desires **recognition** for harvest.*	• ***Communication network** within the community*	• *Focused on **maximizing use** of food.* • ***User-friendly** sharing hub* • *Hopes to **recoup economic losses.***	• ***Green & Sustainable** Lifestyle* • *Regain lost **economic & environmental value***

IDEATION

Definition

Surplus fresh produce
*Because fresh produce is highly perishable, **overstocking or plan changes** often result in waste.*

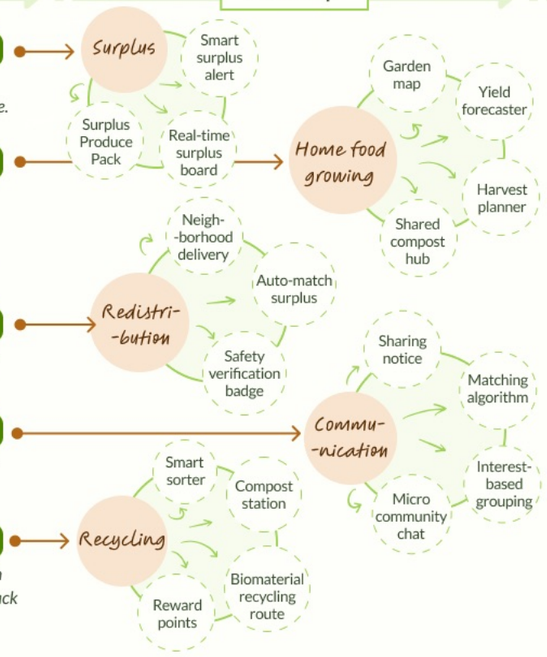
Home food growing
***The cost** of growing vegetables **exceeds** the cost of buying them.*
*Home-grown foods often **mature all at once**, creating more than the household can consume.*

Food Redistribution
*Food redistribution is constrained by **food-safety worries and trust issues.***

Communication network
*Food redistribution is constrained by **food-safety worries and trust issues.***

Poor Recycling Method
The recycling and reuse of surplus produce is often ineffective due to limited material volume and a lack of proper methods.

Mind map



HMW

- Surplus Produce Pack
- Harvest planner
- Neighborhood delivery
- Biomaterial recycling route
- Expiry alert
- Surplus board
- Garden map
- Yield forecaster
- Auto-match algorithm
- Safety verification badge
- Community grouping
- Smart sorter
- Shared compost station
- Materials and tools

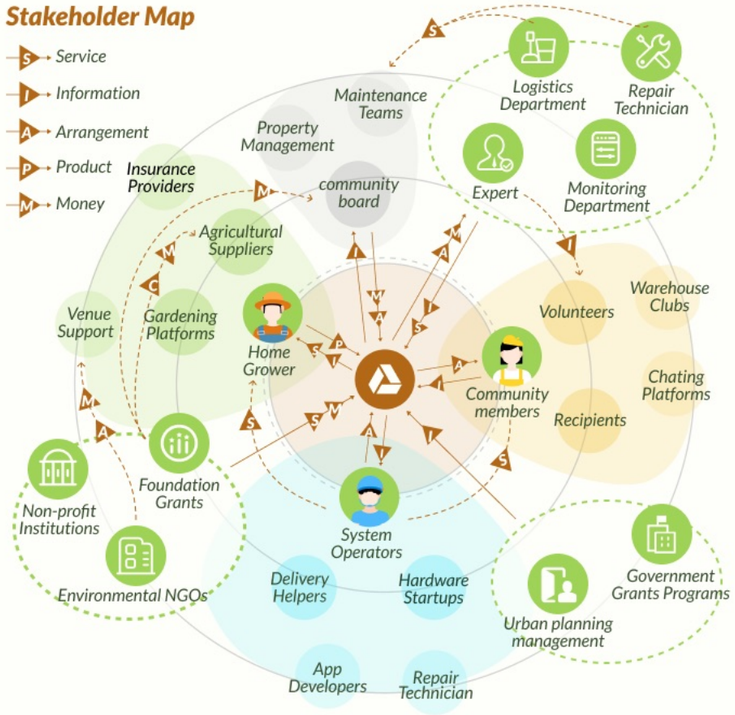
Methods

Service Design
01 **Delivery services** of surplus produce.
02 **Track the origin and flow** of produce.
03 **Staff or volunteers** maintain the process and earn reward points.

APP Design
01 **Display details** about **surplus produce** and **nearby farms.**
02 Establish a produce exchange and recycling **network**
03 Use **computer vision and smart monitoring** to optimize the process

Product Design
01 Modularized and easy-to-use **compost unit** for community food waste
02 Converts surplus food into **eco-friendly material packaging** for system reuse.

Stakeholder Map



CONCEPT



Food Utilization Efficiency

*Optimize community food **distribution** to **reduce waste and maximize value.***



Mobile App

*Build a **platform** to enhance **communication and coordination.***

Sustainable Recycling

*Promote sustainable methods to **recover and repurpose surplus food.***



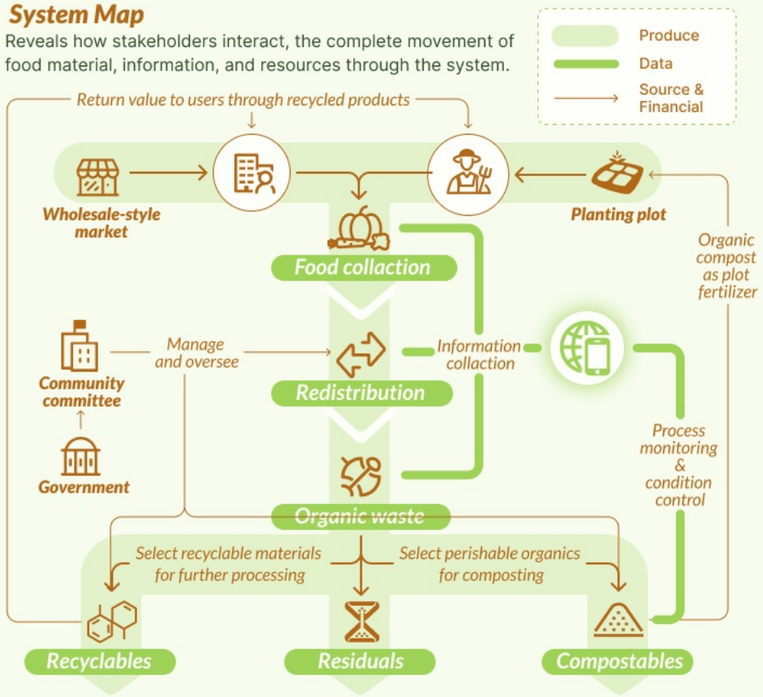
IoT System

*Use smart **monitoring and connectivity** to optimize system **operations and management.***



System Map

Reveals how stakeholders interact, the complete movement of food material, information, and resources through the system.



Desktop Drill

I used print or hand-drawn elements, combined with collage and text, to illustrate the system process.



*Through **paper-based scenario models**, the system can **rapidly and cost-effectively validated** to determine if its processes are **reasonable and feasible.***

FEASIBILITY TEST

To assess the feasibility of these methods in the Freshloop system, I studied and conducted demo trials exploring machine learning, composting, and bio-material reuse.

Bio-based Materials Recycling

I collected daily household food scraps—**fruit peels, cabbage leaves, yam, potato peels, and eggshells**—and tested ways to turn them into reusable bio-materials.



Collect food waste



Chop and process



Dry and grind

*[Using orange peels as an example] Iorange peel, glycerin, gelatin, sodium alginate, and water were **mixed and heated**, then poured into molds and cooled to set.*



The resulting material is tough, elastic and has fragrance, suitable for everyday items.

Composting

I choose to try aerobic composting: **mixing organic waste with dry leaves and branches** under proper conditions.



Raw materials





1. Prepareing



2. Chop them



3. Add nitrogen/ carbon layer



4. Repeat



5. keep moist



6. Completed

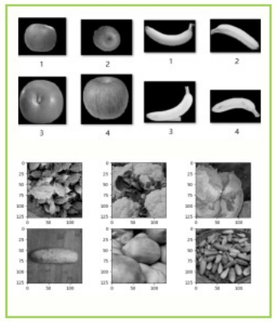
Other attempts



*This method is **simple, low-cost, odor-controlled, and easy to replicate** for long-term community use.*

Machine Learning

To test the method of identifying produce types and quantities, I downloaded and learned an **open-source code from CSDN**; the code primarily uses TensorFlow and CNN.



Part of the recognition code

Data set



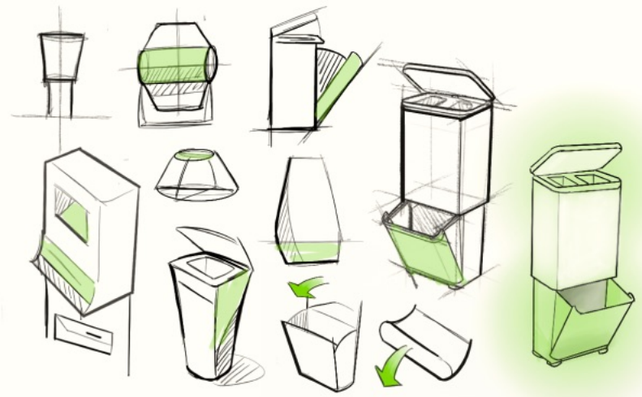
Recognition test

*This demonstrates the feasibility of **integrating intelligent recognition into the App** through further testing.*

PRODUCT DESIGN

Equipment for composting

Sketch



Model

**Unit: Millimeters*



About the Shape

To adapt to community units of different sizes and resident populations, I designed the composting system in a **modular** way. Each composting unit can serve dozens of households, and the **rectangular form allows for more flexible modular placement.**

USER FLOW

Why Use App?

The app provides **mobility, convenience, and real-time updates**, ideal for scattered, continuously generated users' activities, **enabling efficient matching and management**.

Core Function

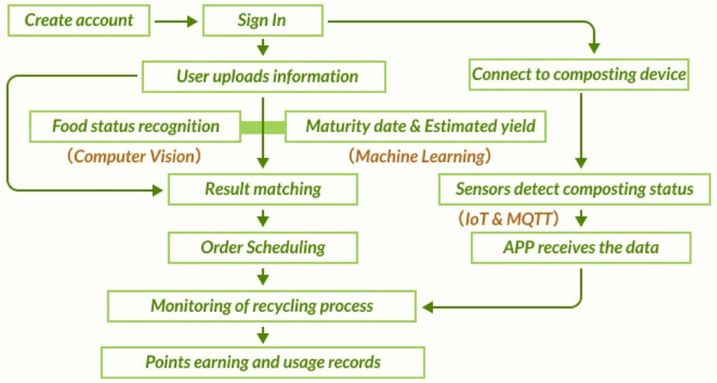
1. Upload surplus
2. Recycling Monitoring
3. Planting Planning
4. Community Sharing

Information Architecture

- **Personalization**
  - **User Profile**
    - Basic Info
      - Reputation & Badges
    - Preferences
      - Dietary Preferences
    - Points & Rewards
      - Sharing / Receiving History
  - **Account Settings**
    - App Preferences
      - Theme mode
    - Account Management
- **Explore**
  - **Urban Farm Map**
    - Map View
      - Location markers
    - Garden Detail Page
      - List view
      - Produce information
      - Plant plot type
      - Harvest Alert Settings
  - **Surplus Produce**
    - List View
      - Food item entry
    - Food Detail Page
      - Contact provider
      - Distance and pickup time
      - Delivery method
- **Redistribution center**
  - **Produce Recognition**
    - CV Analysis Result
      - Upload Image
      - Edit / Confirm Info
  - **Chat & Handover Coordination**
    - Details & Contact
      - Confirm Time & Location
      - Asking for derails
    - Communication
      - Reservation system
      - Request / Offer Interaction
  - **Order & Feedback**
    - Order confirm
      - Select delivery method
      - Evaluation and Feedback
    - Acquire reward points
- **plan & manage**
  - **Harvest Planner**
    - Crop planting tips
      - Growth & harvest timeline
      - Automated yield prediction
  - **Recycling Monitoring**
    - Recycling Progress
      - Destination Tracking
      - Recycled Items Pickup
- **Community**
  - **Sharing**
    - Forum Communication
      - Knowledge & Equipment Sharing
    - Community Trust
      - User Reputation Score
      - Local Home Garden Certification

Data Flow

Clarify the data flow framework of the system combining the App and IoT.



Flow Map & Wireframe



HIFI INTERFACE

UI Visual Specification

01 Color

Primary

Green	#824092

Secondary

Black	#824092	Yellow	#824092	Red	#824092

Neutral

Grey	#824092

02 Component

Placeholder
Placeholder

Placeholder
Placeholder

1
2
3

All(22) Orders(15)
All(22) Orders(15)
All(22) Orders(15)

Continue
Continue

Temperature 11-12-2025 01:00 PM
Humidity 12-12-2025 6:35 AM

03 Logo

FreshLoop

04 Typography

Aa Primary Font **Raleway**

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz
0123456789

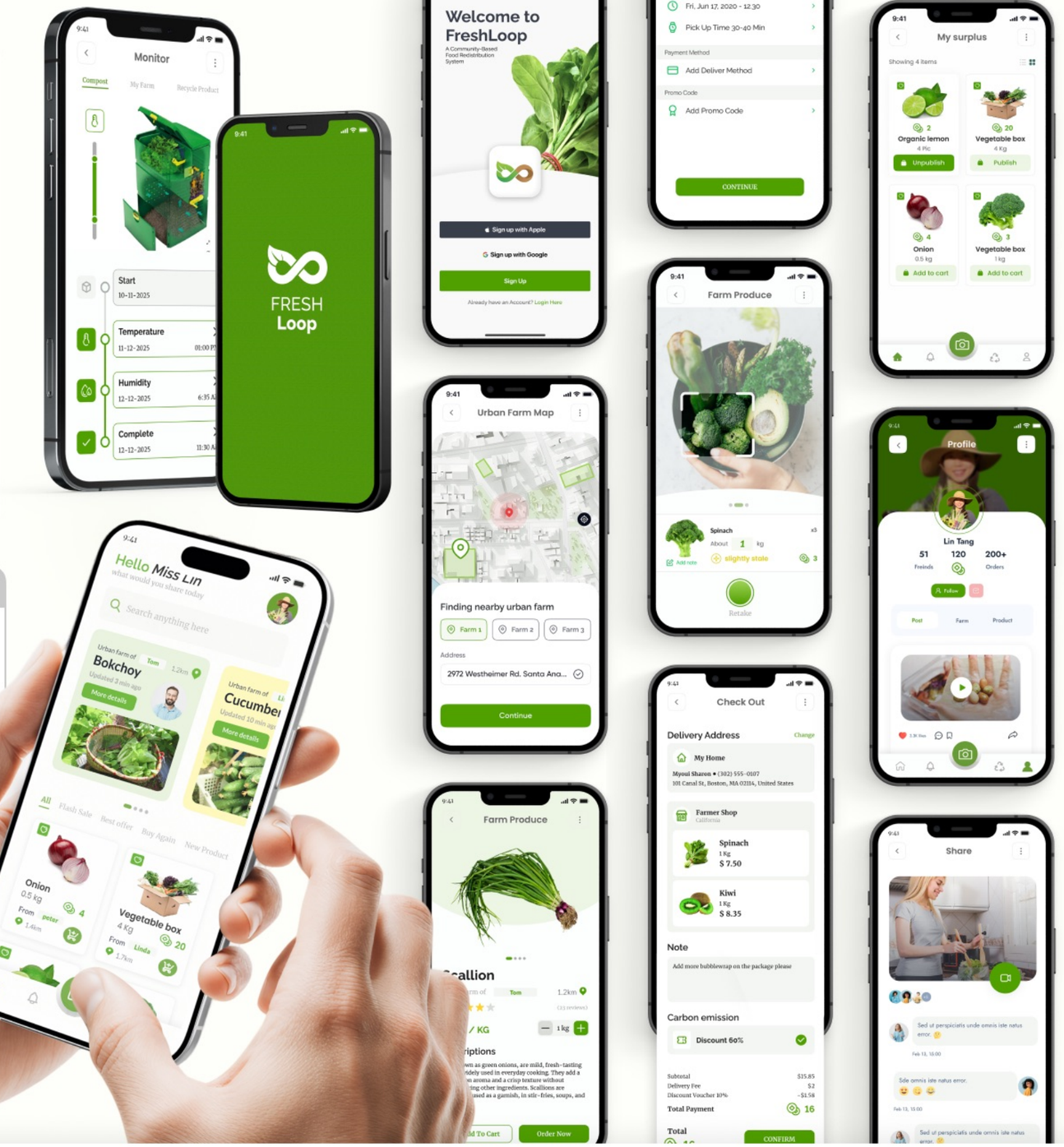
Aa Secondary Font **Merriweather**

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz
0123456789

Heading 1	**Heading 1**	Heading 1
Heading 2	**Heading 2**	Heading 2
Heading 3	**Heading 3**	Heading 3

Paragraph

Body 1	Body 1	**Body 1**	BUTTON	BUTTON	**BUTTON**
Body 2	Body 2	**Body 2**	Caption 1	Caption 1	**Caption 1**
Body 3	Body 3	**Body 3**	Caption 2	Caption 2	**Caption 2**



TESTING

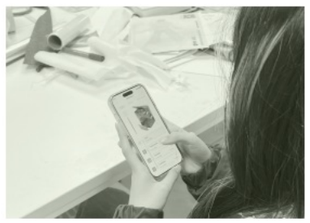
Test Methods

To make the app **functional and user-friendly**, I use **usability testing** to find problems and **A/B testing** to let data choose the best solution.

Tools

ProtoPie was used to prototype app interactions, enabling participants to experience the interface and provide feedback for improving the layout.





Feedback

> *I think the photo recognition interface does not need an additional transition to a detail page; selections can be highlighted directly within the image.*

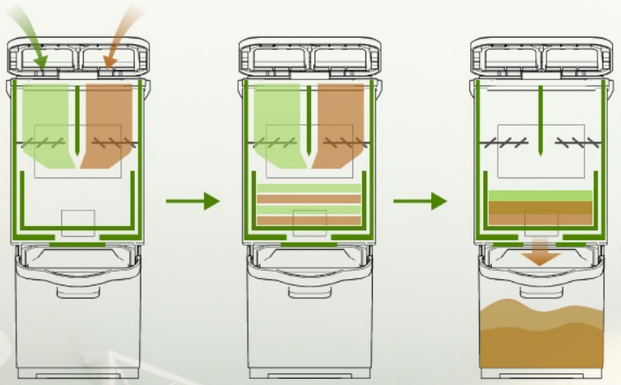
> *The sliding interaction of the recommended tags on the homepage should include snap-to-position behavior.*

> *The selected farm on the map should be clearly indicated with a marker.*



Senario

Compost Operation



Add and shred waste materials

Mix and stir thoroughly

Monitor in real time and obtain compost.

Key indicators

Temperature
40–65°C
To indicate active aerobic decomposition.

Moisture
50–60%
To support microbial activity, prevent anaerobic conditions.

Oxygen
Required to maintain aerobic decomposition and reduce odor formation.

Weight
Declare 30–60%
Indicates the effectiveness of composting.



SERVICE BLUREPRINT

Phase	Discover & Join	Upload, Reserve & Receive	Waste Collection & Recycling	Loop Completion
Evicence	• Social media • Sign-up Interface • Emails & notice	• App interface & notification • Order/pick-up slip • Delivery staff & volunteers	• Collection staff & collection point • Recycling logs • In-app points	• Products • Photos • Redeemable points card
User Journey	Decided to join Registers account Browse apps	Take photos to upload the info Fill in the details Pack up produce Send it to delivery Search for needs Checks status Complete pick-up Receive goods	Collect the waste and delivers them to collection point Scan/deposit & confirm View & accumulate green points	Receive redeemable points Redeem recycled resources Use Products or resources Share results Write reviews
Line of interaction				
Frontstage	Platform registration interface Onboarding guide and education	Posting interface Reservation interface Recommend for user Matching & notification system Delivery process display	Recycling equipment & signage Deposit scanner App monitor interface Points center	"Loop" redemption store App interface Resource and product distribution delivery Showcase and review App interface
Line of visibility				
Backstage	User data verification & database entry Initialize user preference settings	Dispatch tasks to couriers/systems Real-time supply-demand matching algorithm	Real-time composting monitoring The process and proportion calculation of waste recycling Points calculation	Delivery arrangement Update data in real-time
Line of Internal interaction				
Support Processes	• Community governance & rule maintenance • User education content development	• Algorithm optimization & data analysis • Partner relationship management (logistics, insurance) • points settlement system maintenance	• Regular facility maintenance & cleaning • Environmental compliance & government reporting	• Recycled resource supply chain management • Product quality & safety test • Community advocacy & circular outcome marketing

REFLECTION

The FreshLoop service design has begun to take shape—but can it truly be implemented, and how might this be achieved?





For the design to be truly realized, additional research is required on community-specific distances and population density. More thoughts about volunteer delivery operations and the real-world feasibility of an odorless and harmless composting system are needed!